

TagAlert

Is there an AirTag following me?
I need to know now.

Elizabeth Hadik-Barkoczy
University of Arizona | APCV 498 Capstone
Field-tested on an iPhone SE, November 1 to 9, 2026

A planted tag can follow you for hours

Made for keys, used on people

An AirTag is the size of a coin. Hidden in a bag or car, it shows someone else where you are.

The warning can take a day

A tag separated from its owner plays a sound at a random time inside an 8 to 24 hour window.

8-24 h

before a separated AirTag makes a sound (MacTech, 2021)

TagAlert is built for that window: open it, and know in minutes.

Three goals and one rule

1 Find the tag

Detect AirTags near an iPhone, even though iOS hides them.

2 Judge it

Tell a tag that is following you from one that is just nearby.

3 Locate it

Make the tag beep so you can physically find it, safely.

! Measure everything

Every number in this talk comes from a test log.

iOS hides the signal, so the app asks the device

1

Queue

Every unnamed device, strongest signal first.

2

Connect

A brief connection: 6 s, or 10 s if weak.

3

Ask

Does it offer the AirTag service?

4

Decide

Yes: AirTag. Then disconnect, no pairing.

The marker

Service UUID 7DFC9000-7D1C-4951-86AA-8D9728F8D66C. Zero Apple Find My data reached the app in any scan.

Method based on published AirGuard research (SEEMOO Lab, TU Darmstadt). Code written independently in Swift.

10.4 s

median to confirm a tag (21 confirmations)

Nearby is not the same as following

Low	Medium	High
under 1 min, or not moving with you	1 to 2 min, moving with you	2 min or more, moving with you

Time since the tag was first seen

Moving with you

50 m walked (or 60 s in a vehicle) while the tag was seen in 60% of scans, and seen in the last 25 s.

No location

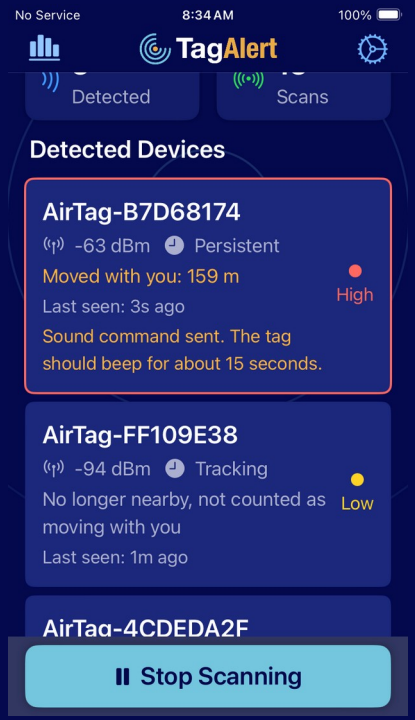
Movement comes from the step counter and activity type. Nothing leaves the phone.

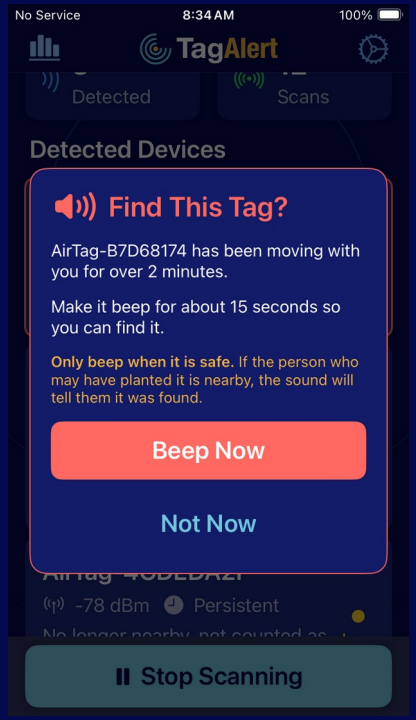
No signal strength

Two tags side by side read about 15 dB apart, so signal is not used.

Why it matters Unknown AirTags turned up on 4 of 7 test walks through an ordinary neighborhood.

The app in the final walk





What it does

- Confirms AirTags in about 10 s
- Low, Medium, High from time and movement
- Notifies at Medium and High
- Asks before beeping the tag
- Safety and reporting guide
- CSV log behind every result

Walk B7, build 6b, November 9, 2026

Results: measured, not estimated

10.4 s

Median time to confirm

21 confirmations, 2.9 to 28 s

123 s

To High in the final walk

148 m walked; Medium at 72 s, 77 m

0

Alerts, tags not moving

7 stationary tests, up to 47 min

26 / 27

Beeps accepted, my tags

separated from their owner

0 / 3

Beeps, owner nearby

same tag: 1 / 1 with owner away

17% / h

Battery, screen on

13% in a 47-minute run

Key finding Apple blocks the beep when the owner is nearby, so it works exactly in the planted-tag case.

What went wrong, and how it got fixed

The pocket tag waited 94 s in line behind distant devices.	➡	**Strongest signal first: confirmed in 6 s.**
A parked tag passed twice on a loop was flagged as following.	➡	**A tag must show up in most scans, and recently.**
Two tags side by side read about 15 dB apart.	➡	**Signal strength removed from the rules.**

28 builds, every change tied to a log. After the fixes: four walks, zero false alerts.

Limits and next steps

Limits

- Works only while the app is open
- Can't see if a tag is with its owner
- One phone, one home, one route
- About 17% battery per hour, screen on

Next

- Android or a companion device to read the owner flag
- Varied routes to tune alert times
- Vehicle and GPS reference tests
- A larger study with real users

Code, logs and report github.com/Hadik-UofA/TagAlert | hadik-uofa.github.io/TagAlert



AirTags are great for finding your keys.
TagAlert is great for when you're the one being found.

Thank you. Questions?

References

Heinrich, Bittner & Hollick (2022). AirGuard. ACM WiSec. doi:10.1145/3507657.3528546

Heinrich, Stute, Kornhuber & Hollick (2021). Who Can Find My Devices? PoPETs 2021(3). doi:10.2478/popets-2021-0045

IETF (2026). Detecting Unwanted Location Trackers Accessory Protocol. draft-ietf-dult-accessory-protocol-01

MacTech (2021, June 3). Apple is beefing up its AirTags privacy features